

Yash Agarwal

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EDUCATION

Vellore Institute of Technology, Vellore | B.Tech CSE with AI & ML Specialization| CGPA: 8.78 2023-Present
Narayana Junior College, Hyderabad | XII Percentage: 90% 2021-2023
Jubilee Hills Public School, Hyderabad | X Percentage: 86% 2009-2021

EXPERIENCE

Data Engineering Intern| Jio Platform

Mumbai | May 2026 - July 2026

- Engineered Kafka-based event streaming pipelines for real-time data replication across 5 topics, processing 50,000+ events/sec in a distributed ingestion architecture
- Designed and validated event-driven data ingestion workflows, Reduced data replication lag, pipeline downtime, and stale-read rate by 30% through Kafka consumer group tuning and dynamic partition rebalancing across 5 topics
- Instrumented pipeline observability using Prometheus and Grafana dashboards, enabling real-time alerting on consumer lag across 10 active Kafka data streams and reducing mean detection time for throughput degradation by 25%

SKILLS

- Programming Languages:** Python, SQL, Java, C++, Object-Oriented Programming (OOP)
- Data, AI & Machine Learning:** Pandas, NumPy, Scikit-learn, TensorFlow, Keras, Matplotlib, Data Pre-processing, Model Training & Evaluation, Linear & Logistic Regression, Neural Networks, Reinforcement Learning (DQN), Generative AI, Apache Kafka, Redis, MySQL, PostgreSQL
- Web Development & Testing:** React.js, HTML5, CSS3, JavaScript (ES6), Flask, Node.js, RESTful APIs, FastAPI, pytest, CI/CD, Unit Testing, Integrations Testing.
- DevOps & Tools:** Git, GitHub, VS Code, [GitHub Actions](#)
- Developer Tools & Practices:** Agile, Software Development Life Cycle (SDLC), Technical Problem-Solving, Docker, Event-Driven Architecture

Technical Projects

Distributed AI Inference Platform | Python, FastAPI, Docker, Redis, Kafka, Prometheus, Grafana |

- Architected a distributed AI inference platform across 12 Dockerized microservices, implementing async request queuing via FastAPI + Redis with Kafka-based task routing, achieving 50 req/sec throughput at 50ms p99 latency under load.
- Built an adaptive Q-Learning scheduler for dynamic batch optimization and request routing, alongside circuit breakers, heartbeat-based health monitoring, and intelligent worker selection
- Integrated production-grade infrastructure including request deduplication, rate limiting, distributed caching, Prometheus metrics, Grafana dashboards, and load testing to demonstrate scalable inference system design.

Autonomous Vehicle Steering Model| Python, TensorFlow, Keras, Pygame, Reinforcement Learning |

- Developed a Deep Q-Network (DQN) agent capable of autonomous navigation within a simulated environment using LiDAR sensor data.
- Designed a complex reward function to optimize for lane keeping and collision avoidance, achieving a 98% successful navigation rate in test epochs.

Distributed Sports Prediction & Wagering Platform | Python, FastAPI, React, MySQL, SQLAlchemy, JWT |

- Built a distributed full-stack event prediction platform that automated live match ingestion, asynchronous bet processing, and leaderboard computation using FastAPI, React, MySQL, and APScheduler, reducing manual intervention through scheduled data synchronization.
- Implemented ACID-compliant transactional workflows using SQLAlchemy row-level locking, JWT authentication, bcrypt hashing, and API rate limiting to maintain consistency and mitigate race conditions during concurrent wallet updates.
- Designed a scalable REST architecture with React Query caching, Zustand state management, Alembic database migrations, and automated administrative integrity checks for secure, high-performance application deployment.

CERTIFICATIONS

- IBM Certified Specialist - Generative AI**
- Supervised Machine Learning: Regression and Classification** - DeepLearning.AI & Stanford University
- Advanced Learning Algorithms** - DeepLearning.AI & Stanford University